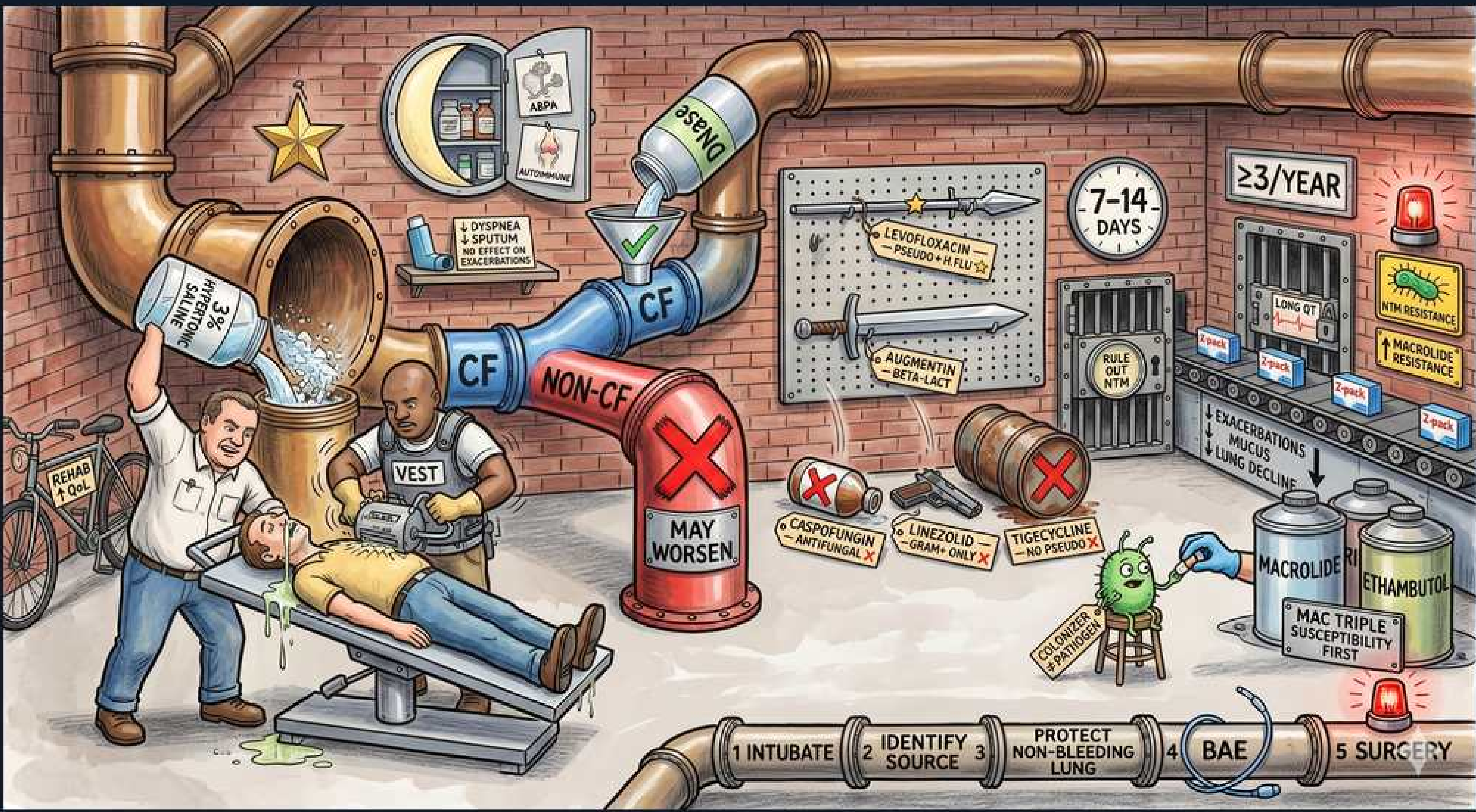
Bronchiectasis — Management, Exacerbations & Hemoptysis



1. Airway clearance (saline pour)

WHAT YOU SEE

A workman tipping a bucket of 'SALINE 3% / 7%' into the top of a big copper pipe — pouring hypertonic saline to flush the airway 'plumbing' = first-line secretion clearance before any drugs.

WHAT IT ENCODES

The cornerstone of all bronchiectasis management is daily airway/secretion clearance, done BEFORE escalating to drugs. This combines chest physiotherapy (active cycle of breathing, postural drainage) with nebulized hypertonic saline 3-7%, which osmotically draws water into the airway lumen to hydrate and loosen tenacious sputum. A short-acting bronchodilator is often given before saline because hypertonic saline can trigger bronchospasm. Mnemonic: 'clear the pipes before you medicate them.'

BOARD TRAP

A patient with daily purulent sputum is started on chronic oral antibiotics without any airway-clearance regimen — WRONG. Mucus clearance (physio + hypertonic saline) is first-line and must be optimized first; chronic antibiotics are reserved for frequent exacerbators despite good clearance. The discriminator is that escalation in bronchiectasis is stepwise, clearance → then antibiotics.

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2. VEST device

WHAT YOU SEE

A second workman wearing an inflatable VEST tending the patient lying on a tilted drainage table — the oscillating vest mechanically shakes secretions loose (high-frequency chest wall oscillation).

WHAT IT ENCODES

High-frequency chest wall oscillation (the inflatable VEST) mechanically vibrates the chest at 5-25 Hz to mobilize peripheral secretions toward the central airways for expectoration. It is one of several adjuncts (oscillatory PEP devices like Flutter/Acapella, autogenic drainage) used when manual physiotherapy alone is insufficient or for self-administered daily clearance. These devices improve sputum yield and quality of life but have not been shown to change mortality.

BOARD TRAP

A board stem may imply a mechanical device REPLACES the need for an active patient clearance routine — WRONG. Devices supplement, not substitute for, a structured daily clearance program; adherence to the overall regimen is what reduces exacerbations, not any single gadget.

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3. Spring / mucolytic valve

WHAT YOU SEE

A coiled SPRING with a green check at the CF pipe junction — the spring 'unkinks' thick CF mucus, representing dornase alfa (DNase), which is CF-specific.

WHAT IT ENCODES

In cystic fibrosis bronchiectasis, recombinant human DNase (dornase alfa, 2.5 mg nebulized daily) cleaves the extracellular DNA released from degraded neutrophils, dramatically lowering CF sputum viscosity and improving FEV1. It is a proven, guideline-recommended CF therapy. Crucially its benefit is CF-SPECIFIC and does not transfer to non-CF bronchiectasis.

BOARD TRAP

A non-CF bronchiectasis patient with thick sputum is prescribed nebulized dornase alfa — WRONG and potentially harmful. RCTs showed DNase WORSENS outcomes (more exacerbations, faster FEV1 decline) in non-CF bronchiectasis. The discriminator: dornase is for CF only; in non-CF use hypertonic saline for mucus, never DNase.

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4. CF vs NON-CF split pipe

WHAT YOU SEE

A blue pipe labelled CF forking into a red branch labelled NON-CF — the pipe literally splits to show management diverges by etiology.

WHAT IT ENCODES

Management forks fundamentally by etiology because the evidence bases differ. Every new bronchiectasis patient needs a workup for an underlying cause (CF sweat chloride/genetics, immunoglobulins for CVID, ABPA testing, alpha-1 antitrypsin, NTM cultures, rheumatologic and ciliary causes) since identifying CF, immunodeficiency, or ABPA changes therapy entirely. CF and non-CF differ in their response to mucolytics, inhaled antibiotics, and the spectrum of colonizing organisms.

BOARD TRAP

Applying a positive CF trial result (e.g., DNase, inhaled tobramycin benefit) directly to a non-CF patient — WRONG. CF trial data do not generalize; non-CF bronchiectasis is a distinct entity. Always anchor the answer to the patient's specific etiology rather than CF dogma.

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5. MAY WORSEN cylinder (non-CF)

WHAT YOU SEE

A red gas cylinder stamped with an X and 'MAY WORSEN' hanging off the NON-CF branch — warns that CF mucolytics (DNase, mannitol) can harm non-CF patients.

WHAT IT ENCODES

Several interventions effective or neutral in CF can be harmful or unproven in non-CF bronchiectasis: dornase alfa worsens it, and inhaled mannitol (an osmotic dry-powder mucolytic) failed to show clear benefit and can provoke bronchospasm. The teaching point is therapeutic humility — do not assume a CF mucoactive agent helps non-CF disease.

BOARD TRAP

Choosing inhaled mannitol or DNase as the 'best mucolytic' for non-CF bronchiectasis — WRONG. In non-CF disease the mucoactive of choice is nebulized hypertonic saline; mannitol/DNase are CF-context answers used here as distractors.

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6. Levofloxacin / Cipro tool

WHAT YOU SEE

A sword/tool on the pegboard labelled LEVOFLOXACIN / CIPRO — the antipseudomonal fluoroquinolone 'weapon' for Pseudomonas-positive exacerbations.

WHAT IT ENCODES

When Pseudomonas aeruginosa is the colonizer/pathogen, the oral exacerbation antibiotic is an antipseudomonal fluoroquinolone: ciprofloxacin 500-750 mg PO BID or levofloxacin 750 mg daily, for 14 days. Ciprofloxacin is essentially the only reliable oral antipseudomonal agent. First isolation of Pseudomonas warrants an eradication attempt (e.g., oral cipro plus inhaled antipseudomonal) because chronic Pseudomonas colonization predicts worse lung function and more exacerbations.

BOARD TRAP

Treating a Pseudomonas-positive bronchiectasis exacerbation with amoxicillin or azithromycin — WRONG, neither covers Pseudomonas. Sputum culture must guide therapy; if Pseudomonas grows, the answer is an antipseudomonal fluoroquinolone (or IV beta-lactam like piperacillin-tazobactam/ceftazidime/cefepime if oral options fail).

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7. Augmentin tool

WHAT YOU SEE

A pegboard tool labelled AUGMENTIN — amoxicillin-clavulanate, the weapon for non-Pseudomonas (H. influenzae) exacerbations.

WHAT IT ENCODES

For non-Pseudomonas exacerbations — most commonly Haemophilus influenzae, also Streptococcus pneumoniae and Moraxella — amoxicillin-clavulanate (e.g., 875/125 mg PO BID) is appropriate empiric therapy, again for ~14 days. Always send sputum culture first and then narrow/tailor to the isolate and prior microbiology, since many patients have a known chronic colonizer that dictates the empiric choice.

BOARD TRAP

A stem with a prior Pseudomonas-positive culture is treated empirically with Augmentin during a new exacerbation — WRONG. Past microbiology predicts current pathogens; a known Pseudomonas colonizer needs antipseudomonal cover even before the new culture returns. Augmentin is correct only when H. influenzae / non-Pseudomonas organisms are expected.

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8. 7-14 DAYS clock

WHAT YOU SEE

A wall clock labelled '7-14 DAYS' — the prolonged antibiotic course for an exacerbation, longer than the 5 days used for ordinary pneumonia.

WHAT IT ENCODES

Acute bronchiectasis exacerbations are treated with a PROLONGED antibiotic course — roughly 14 days — substantially longer than the 5-7 days used for uncomplicated community-acquired pneumonia. The longer duration reflects the structurally damaged, chronically infected airways that clear organisms poorly. An exacerbation is defined clinically by sustained worsening (≥48 h) of cough, sputum volume/purulence, dyspnea, or systemic symptoms, not merely a positive culture.

BOARD TRAP

Prescribing a 5-day course for a bronchiectasis exacerbation 'like for CAP' — WRONG. Five days is too short; treat ~14 days. The discriminator is that bronchiectasis is a chronic suppurative airway disease requiring extended courses to control bacterial load.

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9. ≥3/year + chronic abx

WHAT YOU SEE

A red alarm light over a sign reading '≥3/YEAR' with 'CONSIDER LONG-TERM ANTIBIOTICS' — three-plus exacerbations a year is the trigger to escalate to chronic suppressive antibiotics.

WHAT IT ENCODES

The trigger to escalate to long-term suppressive antibiotics is ≥3 exacerbations per year despite optimized airway clearance. Options are chronic oral macrolide (anti-inflammatory + antimicrobial) or, particularly in Pseudomonas-colonized patients, inhaled antibiotics such as nebulized colistin or tobramycin. The goal is to reduce exacerbation frequency, sputum burden, and lung-function decline.

BOARD TRAP

Starting long-term antibiotics in a patient with only one exacerbation a year, or before clearance is optimized — WRONG. Chronic antibiotics are reserved for frequent exacerbators (≥3/yr) who are already adherent to clearance; jumping to suppressive therapy prematurely drives resistance without benefit.

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10. Macrolide barrel

WHAT YOU SEE

A wooden barrel labelled MACROLIDE bottom-right — chronic low-dose azithromycin, the most evidence-backed long-term oral antibiotic.

WHAT IT ENCODES

Chronic low-dose azithromycin (e.g., 250-500 mg three times weekly) reduces exacerbation frequency in frequent exacerbators through combined immunomodulatory and antimicrobial effects. Before starting, baseline ECG (QT prolongation risk) and liver enzymes are checked, and audiometry is considered with prolonged use. It is the most evidence-backed long-term oral antibiotic for both CF and non-CF bronchiectasis.

BOARD TRAP

Starting chronic azithromycin without first excluding active NTM/MAC infection — WRONG and dangerous. Macrolide monotherapy in unrecognized MAC selects for macrolide-resistant MAC, crippling future treatment. The discriminator: always send NTM sputum cultures and rule out active MAC BEFORE chronic macrolide monotherapy.

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11. Ethambutol barrel (MAC triple)

WHAT YOU SEE

A wooden barrel labelled ETHAMBUTOL tagged 'MAC TRIPLE SUSCEPTIBILITY FIRST' next to the macrolide barrel — the three-drug MAC regimen (macrolide + ethambutol + rifampin), never a macrolide alone.

WHAT IT ENCODES

Mycobacterium avium complex (MAC) pulmonary disease requires multidrug therapy: a macrolide (azithromycin or clarithromycin) PLUS ethambutol PLUS a rifamycin (rifampin), typically given for 12 months past culture conversion. Ethambutol's classic toxicity is dose-dependent optic neuritis (red-green color vision loss), so baseline and periodic eye exams are mandatory.

BOARD TRAP

Treating proven MAC pulmonary disease with azithromycin alone — WRONG. Macrolide monotherapy induces macrolide resistance and treatment failure; MAC always needs the three-drug regimen (macrolide + ethambutol + rifampin). The macrolide must never stand alone against active mycobacteria.

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12. Hemoptysis crossed barrels

WHAT YOU SEE

Two barrels slashed with a big red X (over 'cephalosporin'/'linezolid'/'tigecycline' tags) — drugs/agents to AVOID, with hemoptysis as the bleeding hazard to respect.

WHAT IT ENCODES

Hemoptysis is a hallmark bronchiectasis complication arising from chronically inflamed, hypertrophied, and tortuous BRONCHIAL ARTERIES (systemic, high-pressure circulation), so bleeding can be brisk. Massive hemoptysis (variably defined as >100-600 mL/24 h, or any volume causing airway compromise or hemodynamic instability) is life-threatening primarily through ASPHYXIATION, not exsanguination.

BOARD TRAP

Reaching for aggressive anticoagulation or assuming the threat is blood loss in massive hemoptysis — WRONG. The lethal mechanism is asphyxiation from blood flooding the airways, so the priority is airway protection, and any anticoagulants/antiplatelets should be held and reversed. Death comes from drowning in blood, not low hemoglobin.

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13. Protect non-bleeding lung

WHAT YOU SEE

A bottom banner reading 'PROTECT NON-BLEEDING LUNG' — the airway maneuver: bleeding side DOWN so blood doesn't drown the healthy lung.

WHAT IT ENCODES

The immediate airway maneuver in massive hemoptysis is to protect the healthy lung: position the patient with the BLEEDING side DOWN (lateral decubitus) so gravity keeps blood in the affected lung and out of the good lung. Secure the airway early — intubate if needed, and selective intubation of the non-bleeding mainstem (or a double-lumen tube) can isolate and protect it.

BOARD TRAP

Placing the patient bleeding-side UP (or sitting them fully upright) — WRONG; this lets blood spill into and drown the healthy lung. The rule is bleeding-side DOWN to confine blood to the already-diseased lung and preserve gas exchange in the good one.

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14. BAE / surgery ladder

WHAT YOU SEE

A numbered banner ladder along the bottom: 1 INTUBATE → 2 IDENTIFY SOURCE → PROTECT LUNG → BAE → 5 SURGERY — the stepwise massive-hemoptysis algorithm (embolization before the OR).

WHAT IT ENCODES

Stepwise management of massive hemoptysis: (1) stabilize/protect airway (intubate, bleeding-side down), (2) localize the source (bronchoscopy and/or CT angiography), (3) achieve definitive control with bronchial artery embolization (BAE) as first-line, and (4) reserve surgery (lobectomy) for BAE failure, recurrence, or non-embolizable sources. Bronchoscopy can also provide temporizing local control.

BOARD TRAP

Sending a patient with massive hemoptysis straight to emergency lobectomy — WRONG as the initial definitive step. Bronchial artery embolization is first-line definitive therapy; surgery is reserved for embolization failure or recurrent bleeding. The discriminator: stabilize the airway, localize, then BAE before considering the OR.
